Ancient Footprints Tell of a Lake That Vanished

From: Popular Mechanics Mined: 2026-10-28 3 min read

Long ago, people walked beside a lake in what is now White Sands National Park in New Mexico. Their footprints, left between 23,500 and 21,700 years ago, survived in the sand. Scientists have studied the ground around the prints to learn about the lost world those walkers knew.

The team that first dated the footprints in 2021 went further. It looked at layers with human footprints, animal tracks, and wheel marks. Dating of **conifer** pollen and **aquatic** seeds sharpened the timeline.

The team says the environment changed completely, from a year-round lake to a dry lake bed, during the height of the ice age. At first, Lake Otero filled much of the **basin**, **conifer** forests grew in the mountains, and temperatures were about 4 degrees Celsius colder than today. Then temperatures rose, the lake shrank, forests declined, and oak and **sagebrush** moved in as **evaporation** increased.

Bennett and Reynolds wrote that people visited repeatedly amid environmental change. Computer models helped explain why footprints appear in bursts around times of change: prints vanished unless changing water levels buried and saved them.

Word Treasure

conifer -- A conifer is a tree that usually keeps its needle-like leaves all year and grows cones.

aquatic -- Aquatic means living in or connected with water.

basin -- A basin is a low area of land where water can collect, like a giant natural bowl.

evaporation -- Evaporation is when water turns into vapor and rises into the air.

sagebrush -- Sagebrush is a dry-land shrub with gray-green leaves that grows where there is little rain.

Background you need

During the ice age, much of the world was colder, and big lakes filled low areas of land called basins.

Footprints usually wash away, but they can survive for a very long time when rising water quickly buries them in mud and sand.

Scientists study the layers of ground, along with things like pollen and seeds found in them, to figure out what the land, plants, and weather were like long ago.

Break it down

WHO: Scientists studying ancient footprints, including Bennett and Reynolds.

WHAT: They studied the ground around human footprints to learn about a lost world beside a vanished lake.

WHERE: At what is now White Sands National Park in New Mexico, around the former Lake Otero.

WHY: To understand how the environment changed completely during the ice age and how people kept visiting through those changes.

Why it matters

Footprints in the ground can show you how the place where you live might have looked completely different long ago.

Test what you remember

1. Where were the ancient footprints found?

- (A) A forest in Boston
- (B) A beach in Greece
- (C) White Sands National Park in New Mexico
- (D) A peat bog in England

2. What did the team study besides the footprints themselves?

- (A) Traffic accidents
- (B) Old guitars and synthesizers
- (C) Layers with animal tracks and wheel marks
- (D) Bones in a bog

3. What helped sharpen the timeline of the footprints?

- (A) Measuring the depth of the sand
- (B) Computer models of baseball games
- (C) Counting the rings of oak trees
- (D) Dating of conifer pollen and aquatic seeds

4. How did the environment change during the height of the ice age, according to the team?

- (A) From a dry desert to a deep ocean
- (B) From a year-round lake to a dry lake bed
- (C) From a forest to a snowy mountain
- (D) From a river to a frozen sea

5. What happened as temperatures rose?

- (A) People stopped visiting the lake
- (B) The lake grew and forests spread
- (C) The lake shrank and forests declined
- (D) The footprints disappeared forever

6. Why do footprints appear in bursts around times of change, according to the computer models?

- (A) People only walked there on sunny days
- (B) The sand was too dry to hold prints
- (C) Prints vanished unless changing water levels buried and saved them
- (D) Animals carried the footprints away

Answer key (try the quiz first!): 1: C 2: C 3: D 4: B 5: C 6: C

Questions to think about

1. The research team: Says the environment changed completely, from a year-round lake to a dry lake bed, during the height of the ice age.

2. Bennett and Reynolds: Wrote that people visited the area repeatedly amid environmental change.

MY THOUGHTS (at least 20 words):